

1 Appendix for Section ??

1.1 Asymptotic Consumption Growth Factors

Proposition 1. We have $\lim_{m_t \rightarrow \infty} \mathbb{E}_t[\mathbf{c}_{t+1}/\mathbf{c}_t] = \mathbf{P}$ and $\lim_{m_t \rightarrow 0} \mathbb{E}_t[\mathbf{c}_{t+1}/\mathbf{c}_t] = \infty$.

Proof for Proposition 1. For consumption growth, as $m \rightarrow 0$ we have:

$$\begin{aligned} \lim_{m_t \rightarrow 0} \mathbb{E}_t \left[\left(\frac{\mathbf{c}(m_{t+1})}{\mathbf{c}(m_t)} \right) \tilde{\mathcal{G}}_{t+1} \right] &> \lim_{m_t \rightarrow 0} \mathbb{E}_t \left[\left(\frac{\underline{\mathbf{c}}(\tilde{\mathcal{R}}_{t+1}\mathbf{a}(m_t) + \boldsymbol{\xi}_{t+1})}{\bar{\kappa}m_t} \right) \tilde{\mathcal{G}}_{t+1} \right] \\ &= \wp \lim_{m_t \rightarrow 0} \mathbb{E}_t \left[\left(\frac{\underline{\mathbf{c}}(\tilde{\mathcal{R}}_{t+1}\mathbf{a}(m_t))}{\bar{\kappa}m_t} \right) \mathcal{G}_{t+1} \right] \\ &\quad + (1 - \wp) \lim_{m_t \rightarrow 0} \mathbb{E}_t \left[\left(\frac{\underline{\mathbf{c}}(\tilde{\mathcal{R}}_{t+1}\mathbf{a}(m_t) + \theta_{t+1}/(1 - \wp))}{\bar{\kappa}m_t} \right) \tilde{\mathcal{G}}_{t+1} \right] \\ &> (1 - \wp) \lim_{m_t \rightarrow 0} \mathbb{E}_t \left[\left(\frac{\underline{\mathbf{c}}(\theta_{t+1}/(1 - \wp))}{\bar{\kappa}m_t} \right) \tilde{\mathcal{G}}_{t+1} \right] \\ &= \infty \end{aligned}$$

where the second-to-last line follows because $\lim_{m_t \rightarrow 0} \mathbb{E}_t \left[\left(\frac{\underline{\mathbf{c}}(\tilde{\mathcal{R}}_{t+1}\mathbf{a}(m_t))}{\bar{\kappa}m_t} \right) \tilde{\mathcal{G}}_{t+1} \right]$ is positive, and the last line follows because the minimum possible realization of θ_{t+1} is $\underline{\theta} > 0$ so the minimum possible value of expected next-period consumption is positive.

Next we establish the limit of the expected consumption growth factor as $m_t \rightarrow \infty$:

$$\lim_{m_t \rightarrow \infty} \mathbb{E}_t[\mathbf{c}_{t+1}/\mathbf{c}_t] = \lim_{m_t \rightarrow \infty} \mathbb{E}_t[\tilde{\mathcal{G}}_{t+1}\mathbf{c}_{t+1}/\mathbf{c}_t].$$

But

$$\mathbb{E}_t[\tilde{\mathcal{G}}_{t+1}\mathbf{c}_{t+1}/\bar{\mathbf{c}}_t] \leq \mathbb{E}_t[\tilde{\mathcal{G}}_{t+1}\mathbf{c}_{t+1}/\mathbf{c}_t] \leq \mathbb{E}_t[\tilde{\mathcal{G}}_{t+1}\bar{\mathbf{c}}_{t+1}/\underline{\mathbf{c}}_t]$$

and

$$\lim_{m_t \rightarrow \infty} \tilde{\mathcal{G}}_{t+1}\underline{\mathbf{c}}(m_{t+1})/\bar{\mathbf{c}}(m_t) = \lim_{m_t \rightarrow \infty} \tilde{\mathcal{G}}_{t+1}\bar{\mathbf{c}}(m_{t+1})/\underline{\mathbf{c}}(m_t) = \lim_{m_t \rightarrow \infty} \tilde{\mathcal{G}}_{t+1}m_{t+1}/m_t,$$

while (for convenience defining $\mathbf{a}(m_t) = m_t - \mathbf{c}(m_t)$),

$$\begin{aligned} \lim_{m_t \rightarrow \infty} \tilde{\mathcal{G}}_{t+1}m_{t+1}/m_t &= \lim_{m_t \rightarrow \infty} \left(\frac{\mathbf{R}\mathbf{a}(m_t) + \tilde{\mathcal{G}}_{t+1}\boldsymbol{\xi}_{t+1}}{m_t} \right) \\ &= (\mathbf{R}\beta)^{1/\gamma} = \mathbf{P} \end{aligned}$$

because $\lim_{m_t \rightarrow \infty} \mathbf{a}'(m) = \mathbf{P}/\mathbf{R}^1$ and $\tilde{\mathcal{G}}_{t+1}\boldsymbol{\xi}_{t+1}/m_t \leq (\mathcal{G}\bar{\psi}\bar{\theta}/(1 - \wp))/m_t$ which goes to zero

$$^1 \lim_{m_t \rightarrow \infty} \mathbf{a}(m_t)/m_t = 1 - \lim_{m_t \rightarrow \infty} \mathbf{c}(m_t)/m_t = 1 - \lim_{m_t \rightarrow \infty} \mathbf{c}'(m_t) = \mathbf{P}/\mathbf{R}.$$

as m_t goes to infinity. Hence we have:

$$\mathbf{P} \leq \lim_{m_t \rightarrow \infty} \mathbb{E}_t[\mathbf{c}_{t+1}/\mathbf{c}_t] \leq \mathbf{P}$$

so as cash goes to infinity, consumption growth approaches its value $\mathbf{P}$ in the perfect foresight model. $\square$

This appendix proves Theorems ??-?? and:

Lemma 1. *If $\check{m}$ and $\hat{m}$ both exist, then $\check{m} \leq \hat{m}$.*

1.2 Existence of Buffer Stock Target

1.2.1 Existence of Individual Buffer Stock Target

Proof of Theorem ??. First, observe that $\mathbb{E}_t[m_{t+1}/m_t] = \frac{\mathbb{E}_t((m_t - c(m_t))\tilde{\mathcal{R}}_{t+1} + \xi_{t+1})}{m_t}$. Note that c is continuous since c is concave on $\mathbb{R}_{++}$ by Lemma ??. Thus for any convergent sequence $\{m_t^j\}_{j=0}^\infty$, with $m_t^j \in \mathbb{R}_{++}$, $(m_t^j - c(m_t^j))\tilde{\mathcal{R}}_{t+1} + \xi_{t+1}$ will be bounded above and below. It follows that, using the Dominated Convergence Theorem, $\mathbb{E}_t[m_{t+1}/m_t]$ will be continuous in m_t .

The remainder of the proof proceeds as follows. To establish Equation (??), we will show (i) that there exists a point $\check{m}_t$ where $\mathbb{E}_t[\check{m}_{t+1}^*/\check{m}_t^*] < 1$ and (ii) a point $\hat{m}$ where $\mathbb{E}_t[\hat{m}_{t+1}/\hat{m}_t] > 1$. By continuity of $\mathbb{E}[m_{t+1}/m_t]$ in m_t and the Intermediate Value Theorem, there will exist $\hat{m}$ such that $\mathbb{E}_t[\hat{m}_{t+1}/\hat{m}_t] = 1$. In turn, to establish that $\hat{m}$ is a point of stability, Equation (??), we will show that (iii) $\mathbb{E}_t[m_{t+1}] - m_t$ is decreasing.

Part (i). Existence of $\check{m}_t$ where $\mathbb{E}_t[\check{m}_{t+1}/\check{m}_t] < 1$.

To proceed, first suppose return impatience holds and take the steps analogous to those leading to Equation (1) in the proof of proof for Proposition 1, but dropping the $\mathcal{G}_{t+1}$ from the RHS:

$$\begin{aligned} \lim_{m_t \rightarrow \infty} \mathbb{E}_t[m_{t+1}/m_t] &= \lim_{m_t \rightarrow \infty} \mathbb{E}_t \left[\frac{\tilde{\mathcal{R}}_{t+1}(m_t - c(m_t)) + \xi_{t+1}}{m_t} \right] \\ &= \mathbb{E}_t[(R/\tilde{\mathcal{G}}_{t+1})\mathbf{P}/R] \\ &= \mathbb{E}_t[\mathbf{P}/\tilde{\mathcal{G}}_{t+1}] \\ &< 1, \end{aligned}$$

where the inequality follows from strong growth impatience. By continuity of $\mathbb{E}_t[m_{t+1}/m_t]$ in m_t , there exists $\check{m}_t$ large enough such that $\mathbb{E}_t[\check{m}_{t+1}/\check{m}_t] < 1$.

Next, suppose return impatience fails. The fact that $\lim_{m_t \rightarrow \infty} \frac{c(m_t)}{m_t} = 0$ (Lemma ??) means the limit of the RHS of (1) as $m_t \rightarrow \infty$ is $\bar{\mathcal{R}} = \mathbb{E}_t[\tilde{\mathcal{R}}_{t+1}]$. Equations (2)-(3) below show that when strong growth impatience holds and return impatience fails $\bar{\mathcal{R}} < 1$.

Thus, we have $\lim_{m \rightarrow \infty} \mathbb{E}[m_{t+1}/m_t] < 1$ whether the return impatience holds or fails.

Part (ii). Existence of $\dot{m}_t > 1$ where $\mathbb{E}_t[\dot{m}_{t+1}/\dot{m}_t] > 1$.

Analogous to Equation (1), the ratio of $\mathbb{E}_t[m_{t+1}]$ to m_t is unbounded above as $m_t \rightarrow 0$ because $\lim_{m_t \rightarrow 0} \mathbb{E}[m_{t+1}] > 0$. Thus, if $\mathbb{E}_t[m_{t+1}/m_t]$ is continuous in m_t , and takes on values above and below one at $\dot{m}_t$ and $\check{m}_t$, by the Intermediate Value Theorem, there must be at least one point at which it is equal to one.

Part (iii). $\mathbb{E}_t[m_{t+1}] - m_t$ is strictly decreasing.

Finally to show $\mathbb{E}_t[m_{t+1}] - m_t$ is strictly decreasing m_t , define $\zeta(m_t) := \mathbb{E}_t[m_{t+1}] - m_t$ and note that:

$$\begin{aligned}\zeta(m_t) < 0 &\leftrightarrow \mathbb{E}_t[m_{t+1}/m_t] < 1 \\ \zeta(m_t) = 0 &\leftrightarrow \mathbb{E}_t[m_{t+1}/m_t] = 1 \\ \zeta(m_t) > 0 &\leftrightarrow \mathbb{E}_t[m_{t+1}/m_t] > 1, ,\end{aligned}$$

so that $\zeta(\hat{m}) = 0$. Our goal is to prove that $\zeta(\bullet)$ is strictly decreasing on $(0, \infty)$. Let Δ_ϵ be the finite forward difference for spacing $\epsilon > 0$. Fixing $\epsilon > 0$, we will have:

$$\begin{aligned}\Delta_\epsilon \zeta(m_t) &= \mathbb{E}_t \left[\Delta_\epsilon \left(\tilde{\mathcal{R}}(m_t - c(m_t)) + \xi_{t+1} - m_t \right) \right] \\ &= \tilde{\mathcal{R}}(\epsilon - \Delta_\epsilon c(m_t)) - \epsilon = \epsilon \left(\tilde{\mathcal{R}} \left[1 - \frac{\Delta_\epsilon c(m_t)}{\epsilon} \right] - 1 \right).\end{aligned}$$

Notice that $\frac{\Delta_\epsilon c(m_t)}{\epsilon} \leq \frac{c(m_t)}{m_t} < 1$ since $\frac{c(m_t)}{m_t}$ is decreasing in m_t by Claim ?? in Appendix ??. Consider the case when return impatience holds. Equation (??) and Lemma ?? indicate $0 < \underline{\kappa} \leq \frac{c(m_t)}{m_t} < 1$. It follows that:

$$\begin{aligned}\tilde{\mathcal{R}} \left[1 - \frac{\Delta_\epsilon c'(m_t)}{\epsilon} \right] - 1 &\leq \tilde{\mathcal{R}}(1 - \underbrace{(1 - \mathbf{P}/R)}_{\underline{\kappa}}) - 1 \\ &= \tilde{\mathcal{R}}\mathbf{P}/R - 1 \\ &= \mathbb{E}_t \left[\frac{R}{\mathcal{G}\psi_{t+1}} \frac{\mathbf{P}}{R} \right] - 1 \\ &= \mathbb{E}_t \left[\frac{\mathbf{P}}{\mathcal{G}\psi_{t+1}} \right] - 1 \\ &\quad \underbrace{= \mathbf{P}/\mathcal{G}\mathbb{E}[\psi^{-1}]}_{\text{by return impatience}}\end{aligned}$$

which is negative because the strong growth impatience says $\mathbf{P}/\mathcal{G}\mathbb{E}[\psi^{-1}] < 1$. Conversely, when return impatience holds fails, recall $\lim_{m_t \rightarrow \infty} \frac{c(m_t)}{m_t} = 0$. This means $\Delta_\epsilon \zeta(m_t)$ from (1) is guaranteed to be negative if:

$$\tilde{\mathcal{R}} = \mathbb{E}_t \left[\frac{R}{\mathcal{G}\psi_{t+1}} \right] < 1. \tag{1}$$

But the combination of the strong growth impatience holding and the return impatience

failing can be written:

$$\overbrace{\mathbb{E}_t \left[\frac{\mathbf{P}}{\mathcal{G}\psi_{t+1}} \right]}^{\mathbf{P}/\mathcal{G}\mathbb{E}[\psi^{-1}]} < 1 < \overbrace{\frac{\mathbf{P}}{\mathbf{R}}}^{\mathbf{P}/\mathbf{R}}, \quad (2)$$

and multiplying all three elements by $\mathbf{R}/\mathbf{P}$ gives:

$$\mathbb{E}_t \left[\frac{\mathbf{R}}{\mathcal{G}\psi_{t+1}} \right] < \mathbf{R}/\mathbf{P} < 1, \quad (3)$$

which satisfies our requirement in (1), thus completing the proof. $\square$

1.2.2 Existence of Pseudo-Steady-State

Proof of Theorem ??. Since by assumption $0 < \underline{\psi} \leq \psi_{t+1} \leq \bar{\psi} < \infty$, our proof in 1.2.1 that demonstrated existence and continuity of $\mathbb{E}[m_{t+1}/m_t]$ implies existence and continuity of $\mathbb{E}[\psi_{t+1}m_{t+1}/m_t]$.

Part (i). Existence of a stable point

Since by assumption $0 < \underline{\psi} \leq \psi_{t+1} \leq \bar{\psi} < \infty$, our proof in Subsection 1.2.1 that the ratio of $\mathbb{E}[m_{t+1}]$ to m_t is unbounded as $m_t \rightarrow 0$ implies that the ratio $\mathbb{E}[\psi_{t+1}m_{t+1}]$ to m_t is unbounded as $m_t \rightarrow 0$. The limit of the expected ratio as $m_t \rightarrow \infty$ goes to infinity is can be found as follows:

$$\begin{aligned} \lim_{m_t \rightarrow \infty} \mathbb{E}_t[\psi_{t+1}m_{t+1}/m_t] &= \lim_{m_t \rightarrow \infty} \mathbb{E}_t \left[\frac{\tilde{\mathcal{G}}_{t+1} \left((\mathbf{R}/\tilde{\mathcal{G}}_{t+1})a(m_t) + \boldsymbol{\xi}_{t+1} \right) / \mathcal{G}}{m_t} \right] \\ &= \lim_{m_t \rightarrow \infty} \mathbb{E}_t \left[\frac{(\mathbf{R}/\mathcal{G})a(m_t) + \psi_{t+1}\boldsymbol{\xi}_{t+1}}{m_t} \right] \\ &= \lim_{m_t \rightarrow \infty} \left[\frac{(\mathbf{R}/\mathcal{G})a(m_t) + 1}{m_t} \right] \\ &= (\mathbf{R}/\mathcal{G})\mathbf{P}/\mathbf{R} \\ &= \mathbf{P}/\mathcal{G} \\ &< 1, \end{aligned}$$

where the last two lines are merely a restatement of growth impatience.

To conclude Part (i) of the proof, the Intermediate Value Theorem says that if $\mathbb{E}[\psi_{t+1}m_{t+1}/m_t]$ is continuous, and takes on values above and below 1, there must be at least one point at which it is equal to one.

Part (ii). $\mathbb{E}_t[\psi_{t+1}m_{t+1}] - m_t$ is monotonically decreasing.

Define $\zeta(m_t) := \mathbb{E}_t[\psi_{t+1}m_{t+1}] - m_t$ and note that:

$$\begin{aligned}\zeta(m_t) < 0 &\leftrightarrow \mathbb{E}_t[\psi_{t+1}m_{t+1}/m_t] < 1 \\ \zeta(m_t) = 0 &\leftrightarrow \mathbb{E}_t[\psi_{t+1}m_{t+1}/m_t] = 1 \\ \zeta(m_t) > 0 &\leftrightarrow \mathbb{E}_t[\psi_{t+1}m_{t+1}/m_t] > 1,\end{aligned}$$

so that $\zeta(\hat{m}) = 0$. Our goal is to prove that $\zeta(\bullet)$ is strictly decreasing on $(0, \infty)$. Letting Δ_ϵ be the forward difference operator, we have:

$$\begin{aligned}\Delta_\epsilon \zeta(m_t) &= \mathbb{E} \left[\Delta_\epsilon \left(\frac{R}{\mathcal{G}}(m_t - c(m_t)) + \psi_{t+1}\mathbf{\xi}_{t+1} - m_t \right) \right] \\ &= \frac{R}{\mathcal{G}}(\epsilon - \Delta_\epsilon c'(m_t)) - \epsilon = \epsilon \left(\frac{R}{\mathcal{G}} \left[1 - \frac{\Delta_\epsilon c(m_t)}{\epsilon} \right] - 1 \right).\end{aligned}$$

for any given $\epsilon > 0$. Notice that $\frac{\Delta_\epsilon c'(m_t)}{\epsilon} \leq \frac{c(m_t)}{m_t} < 1$ since $\frac{c(m_t)}{m_t}$ is decreasing in m_t by Claim ?? in Appendix. Now, we show that $\zeta(m)$ is decreasing when return impatience holds and when return impatience fails. When return impatience holds, Equation (??) and Lemma ?? indicate that $\underline{\kappa} > 0$ and $0 < \underline{\kappa} \leq \frac{c(m_t)}{m_t} < 1$. It follows that:

$$\begin{aligned}\frac{R}{\mathcal{G}}(1 - c'(m_t)) - 1 &< \frac{R}{\mathcal{G}}(1 - \underbrace{(1 - \mathbf{P}/R)}_{\underline{\kappa}}) - 1 \\ &= (R/\mathcal{G})\mathbf{P}/R - 1,\end{aligned}$$

which is negative because growth impatience says $\mathbf{P}/\mathcal{G} < 1$. Conversely, when return impatience holds fails, recall $\lim_{m_t \rightarrow \infty} \frac{c(m_t)}{m_t} = 0$. In turn, this means $\Delta_\epsilon \zeta(m_t)$ from (4) is guaranteed to be negative if:

$$(R/\mathcal{G}) < 1. \tag{4}$$

But we showed in Section ??, Equation (??), that the only circumstances under which the problem has a non-degenerate solution while return impatience fails were ones where the finite limiting human wealth also fails. Thus, $(R/\mathcal{G}) < 1$, completing the proof. $\square$